# Mathematics Bootcamp

## Properties of Preferences and Utility Functions

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# Today

- Preference Relations, Indifference, and Utility
- Monotonicity and Local Nonsatiation
- Convexity and Quasiconcavity
- Continuity Through Contour Sets
- Continuous Utility Representation

# Preference Relations

- A **preference relation** $\succeq$ on $X$ is a subset of $X \times X$.
- $x \succeq y$ reads: $x$ is *at least as good* as y (x is *weakly better* than y).
- From $\succeq$ we derive:
  - Strict preference: $x \succ y \iff (x \succeq y$ and not $y \succeq x)$.
  - Indifference: $x \sim y \iff (x \succeq y$ and $y \succeq x)$.

# Properties of Indifference

## Indifference Relation

If $\succeq$ is rational (i.e., complete and transitive), then the induced indifference relation $\sim$ is:

1. **Reflexive**: $x \sim x$ for all $x \in X$.
2. **Symmetric**: $x \sim y \implies y \sim x$.
3. **Transitive**: $x \sim y \wedge y \sim z \implies x \sim z$.

## Equivalence Relation

An equivalence relation is a binary relation that is reflexive, symmetric and transitive.

# Equivalence Classes

### Equivalence Classes

An equivalence relation induces a **partition** of the set: mutually disjoint subsets whose union equals the entire set.

If $\succeq$ is rational, its induced indifference relation $\sim$

1. is an equivalence relation;
2. generates equivalence classes of indifferent elements.

For each $x \in X$, the equivalence class of $x$ is denoted by

$$[x] = \{y \in X \mid x \sim y\}.$$

# Equivalence Classes and Utility

## Utility representation

A function $u : X \to \mathbb{R}$ *represents* $\succeq$ if

$$x \succeq y \quad \Leftrightarrow \quad u(x) \geq u(y).$$

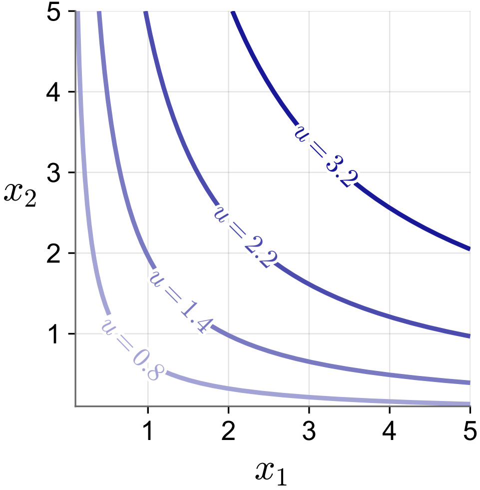
## What the Representation Implies

If $u$ represents $\succeq$, then

$$x \sim y \quad \Leftrightarrow \quad u(x) = u(y).$$

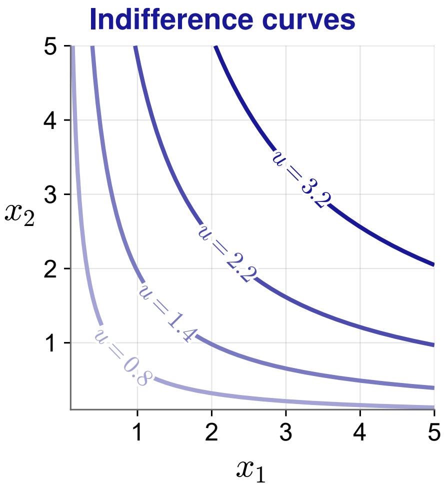
Every equivalence class maps to one utility level; utility is **ordinal**.

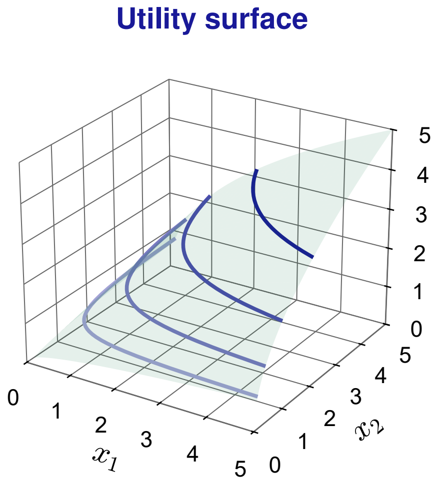
# Indifference Curves




- Indifference curves are equivalence classes of $X$ that map to the same utility in $\mathbb{R}$.
- Equivalence classes are disjoint, so indifference curves can never cross.

# Utility Representation



# Represent Commodities with Ordered Lists

## 2.B Commodities

The decision problem faced by the consumer in a market economy is to choose consumption levels of the various goods and services that are available for purchase in the market. We call these goods and services *commodities*. For simplicity, we assume that the number of commodities is finite and equal to $L$ (indexed by $\ell = 1, \ldots, L$).

As a general matter, a *commodity vector* (or commodity *bundle*) is a list of amounts of the different commodities,

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_L \end{bmatrix},$$

and can be viewed as a point in $\mathbb{R}^L$, the commodity *space*.[1]

We can use commodity vectors to represent an individual's consumption levels. The $\ell$th entry of the commodity vector stands for the amount of commodity $\ell$ consumed. We then refer to the vector as a *consumption vector* or *consumption bundle*.

- $x \geq y$ iff $x_\ell \geq y_\ell$ for every $\ell$; and $x \gg y$ iff $x_\ell > y_\ell$ for every $\ell$.
- So $x \geq y$ with $x \neq y$ means $x_\ell > y_\ell$ for *some* $\ell$ — weaker than $x \gg y$.

# Monotonicity: More is Better

## MWG 3.B.2: Monotonicity

Assume consuming more is always feasible: $x \in X$ and $y \geq x$ imply $y \in X$. The preference relation $\succeq$ on $X$ is

*monotone* if $y \gg x$ implies $y \succ x$;

*strongly monotone* if $y \geq x$ and $y \neq x$ imply $y \succ x$.

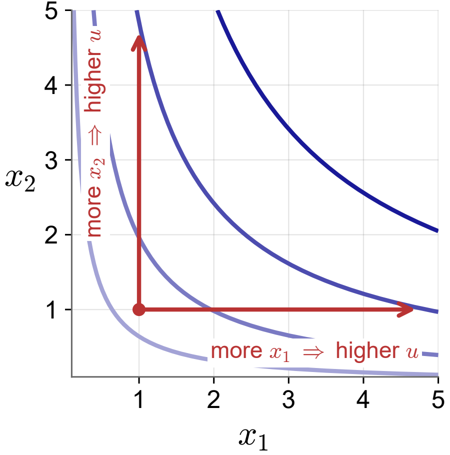
## Proposition

If $u : X \to \mathbb{R}$ represents $\succeq$, then

$\succeq$ is monotone *iff* $u(y) > u(x)$ whenever $y \gg x$;

$\succeq$ is strongly monotone *iff* $u$ is *strictly increasing*: $u(y) > u(x)$ whenever $y \geq x$ and $y \neq x$.

# Monotonicity: More is Better




**Domain:** $X = \mathbb{R}^2_{++}$.

# Local Nonsatiation: No Thick Indifference Curves

**Definition: Local Nonsatiation**

A preference relation $\succeq$ on $X$ is *locally nonsatiated* if for every $x \in X$ and every $\varepsilon > 0$, there exists $y \in X$ such that $\|x - y\| \leq \varepsilon$ and $y \succ x$.

Euclidean distance: $\|x - y\| = \sqrt{\sum_{i=1}^{L}(x_i - y_i)^2}$.

# Local Nonsatiation: No Thick Indifference Curves

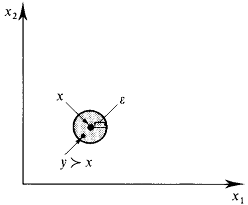



**Figure 3.B.1**
The test for local nonsatiation.

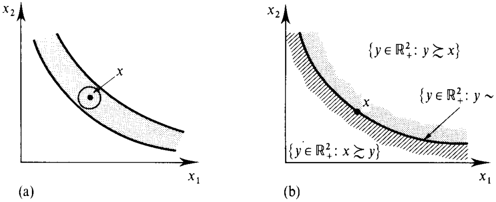



**Figure 3.B.2**
(a) A thick indifference set violates local nonsatiation.
(b) Preferences compatible with local nonsatiation.

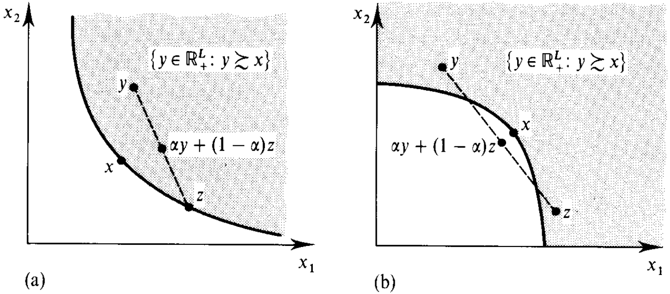
# Monotonicity and Local Nonsatiation

**Proposition**

On $X = \mathbb{R}^L_+$, monotonicity implies local nonsatiation.

However, local nonsatiation does not imply monotonicity.

# Convexity: Taste for Variety




**Definition 3.B.4:** The preference relation $\succsim$ on $X$ is *convex* if for every $x \in X$, the upper contour set $\{y \in X : y \succsim x\}$ is convex; that is, if $y \succsim x$ and $z \succsim x$, then $\alpha y + (1-\alpha)z \succsim x$ for any $\alpha \in [0, 1]$.

Figure 3.B.3(a) depicts a convex upper contour set; Figure 3.B.3(b) shows an upper contour set that is not convex.

# Convexity: Taste for Variety

## Convex Sets and Preferences

A set $X \subseteq \mathbb{R}^L$ is *convex* if $x, y \in X$ and $\alpha \in [0,1]$ imply $\alpha x + (1-\alpha)y \in X$.
Let $X$ be convex and $U(x) := \{y \in X : y \succeq x\}$. Then $\succeq$ is *convex* if

$$y, z \in U(x),\ \alpha \in [0,1] \quad \Longrightarrow \quad \alpha y + (1-\alpha)z \in U(x),$$

and *strictly convex* if that conclusion is $\succ x$ for $y \neq z$, $\alpha \in (0,1)$.

Every upper contour set $U(x)$ is convex.

# Level Sets and Quasiconcavity

## Definitions

Let $X \subseteq \mathbb{R}^L$ be convex. A function $f : X \to \mathbb{R}$ is *quasiconcave* if, for $x, y \in X$ and $\alpha \in [0, 1]$,

$$f(\alpha x + (1 - \alpha)y) \geq \min\{f(x), f(y)\}.$$

It is *strictly quasiconcave* if, for $x \neq y$ and $\alpha \in (0, 1)$,

$$f(\alpha x + (1 - \alpha)y) > \min\{f(x), f(y)\}.$$

**Equivalently.** $f$ is quasiconcave iff every upper level set $\{\, x \in X \mid f(x) \geq c \,\}$ is convex.

# Convex Preferences and Quasiconcave Utility

**Kreps 2.8(b)**

Let $X$ be convex and let $u : X \to \mathbb{R}$ represent $\succeq$. Then

$$\begin{aligned} \succeq \ \text{ is convex} \quad &\iff \quad u \text{ is quasiconcave,} \\ \succeq \ \text{ is strictly convex} \quad &\iff \quad u \text{ is strictly quasiconcave.} \end{aligned}$$

# Properties of a Representation

| Property of $\succeq$ | Property of a representation $u$ |
|---|---|
| Monotonicity | $u(y) > u(x)$ whenever $y \gg x$. |
| Strong monotonicity | $u$ is strictly increasing. |
| Convexity | $u$ is quasiconcave. |
| Strict convexity | $u$ is strictly quasiconcave. |

Continuity is different: that is Debreu's theorem.

# Topology on the Choice Set

### Relative Topology on $X \subseteq \mathbb{R}^L$

For $x \in X$ and $\varepsilon > 0$, the relative open ball is

$$B_\varepsilon^X(x) = \{y \in X : \|y - x\| < \varepsilon\}.$$

- A set $G \subseteq X$ is *open relative to* $X$ if every $x \in G$ has some $\varepsilon > 0$ with $B_\varepsilon^X(x) \subseteq G$.
- A set $F \subseteq X$ is *closed relative to* $X$ if $X \setminus F$ is open relative to $X$.

# Closed Sets Contain Their Limits

## Sequential Test for Closed Sets

Let $F \subseteq X \subseteq \mathbb{R}^L$. The set $F$ is closed relative to $X$ if and only if

$$x_n \in F \text{ for every } n, \qquad x_n \to x \in X \quad \Longrightarrow \quad x \in F.$$

# Lower and Upper Contour Sets

## Contour Sets at $x$

$$U(x) := \{y \in X \mid y \succeq x\} \qquad L(x) := \{y \in X \mid x \succeq y\}.$$

The indifference set through $x$ is always

$$I(x) := \{y \in X : y \sim x\} = U(x) \cap L(x).$$

## Weak and Strict Sections

If $\succeq$ is complete, then

$$\{y \in X : y \succ x\} = X \setminus L(x), \qquad \{y \in X : x \succ y\} = X \setminus U(x).$$

# Lexicographic Preferences

**Example 3.C.1: The Lexicographic Preference Relation**

Let $X = \mathbb{R}^2_+$. Define $x \succeq y$ if either

$$x_1 > y_1, \qquad \text{or} \qquad x_1 = y_1 \text{ and } x_2 \geq y_2.$$

It is complete, transitive, strongly monotone, and strictly convex — yet no utility function represents it.

# Continuity: Limits of Rankings

### MWG 3.C.1: Continuity

Let $X \subseteq \mathbb{R}^L$ and let $\succeq$ be complete. Then $\succeq$ is *continuous* if it is preserved under limits:

$$(x_n, y_n) \to (x, y), \quad x_n \succeq y_n \text{ for every } n \quad \Longrightarrow \quad x \succeq y.$$

# Continuity: Equivalent Tests

### Kreps 1.14: Equivalent Characterizations

For complete, transitive $\succeq$, continuity is equivalent to each of:

1. $U(x)$ and $L(x)$ are closed relative to $X$, for every $x$.
2. $\{y \in X : y \succ x\}$ and $\{y \in X : x \succ y\}$ are open relative to $X$, for every $x$.
3. If $x \succ y$, there is $\varepsilon > 0$ with $x' \succ y'$ whenever $\|x' - x\| < \varepsilon$ and $\|y' - y\| < \varepsilon$.

Kreps takes (3) as the definition; MWG takes the limit form.

# Continuous Representations and Closed Contours

## Closed Level Sets

If $f : X \to \mathbb{R}$ is continuous, then for every $c \in \mathbb{R}$ the sets

$$f^{-1}\big([c, \infty)\big) \qquad \text{and} \qquad f^{-1}\big((-\infty, c]\big)$$

are closed relative to $X$.

A continuous representation therefore implies continuous preferences.

# Debreu's Representation Theorem

**Kreps 2.9: Debreu's Theorem**

Let $X = \mathbb{R}^L_+$. If $\succeq$ is complete, transitive, and continuous on $X$, then there exists a *continuous* utility function $u : X \to \mathbb{R}$ such that

$$x \succeq y \quad \Longleftrightarrow \quad u(x) \geq u(y).$$

- It gives *at least one* continuous representation — not every representation is continuous.

# Three Assumptions, Three Conclusions

| **Assumptions** | **Conclusion** |
|---|---|
| Nonempty compact $D$; continuous $u$ | $\arg\max_{x \in D} u(x)$ is nonempty. |
| Convex $D$; strictly quasiconcave $u$ | There is at most one maximizer. |
| Local nonsatiation; $D = B(p, I)$ with $p \gg 0$ | Every optimum satisfies $p \cdot x = I$. |

Day 4 takes each in turn.